

Assignment No. 1

```
from collections import defaultdict, deque
```

```
class Graph:
```

```
    directed = True
```

```
    def __init__(self):
```

```
        self.graph = defaultdict(list)
```

```
    def addEdge(self, u, v):
```

```
        self.graph[u].append(v)
```

```
        if not self.directed:
```

```
            self.graph[v].append(u)
```

```
    def DFS(self, v, d, visitSet = None) -> bool:
```

```
        visited = visitSet or set()
```

```
        visited.add(v)
```

```
        print(v, end=" ")
```

```
        if v == d:
```

```
            return True
```

```
        for neighbour in self.graph[v]:
```

```
            if neighbour not in visited:
```

```
                if self.DFS(neighbour, d, visited):
```

```
                    return True
```

```
        return False
```

```
    def BFS(self, s, d):
```

```
        visited = defaultdict(bool)
```

```
        queue = deque([s])
```

```
visited[s] = True
```

```
while queue:
```

```
    s = queue.popleft()
```

```
    print (s, end = " ")
```

```
    if s == d:
```

```
        return
```

```
    for i in self.graph[s]:
```

```
        if visited[i] == False:
```

```
            queue.append(i)
```

```
            visited[i] = True
```

```
# A1.png
```

```
if __name__ == '__main__':
```

```
    g = Graph()
```

```
    g.addEdge('H', 'A')
```

```
    g.addEdge('A', 'D')
```

```
    g.addEdge('A', 'B')
```

```
    g.addEdge('B', 'F')
```

```
    g.addEdge('B', 'C')
```

```
    g.addEdge('C', 'E')
```

```
    g.addEdge('C', 'G')
```

```
    g.addEdge('C', 'H')
```

```
    g.addEdge('G', 'H')
```

```
    g.addEdge('G', 'E')
```

```
    g.addEdge('E', 'F')
```

```
    g.addEdge('E', 'B')
```

```
    g.addEdge('F', 'A')
```

```
    g.addEdge('D', 'F')
```

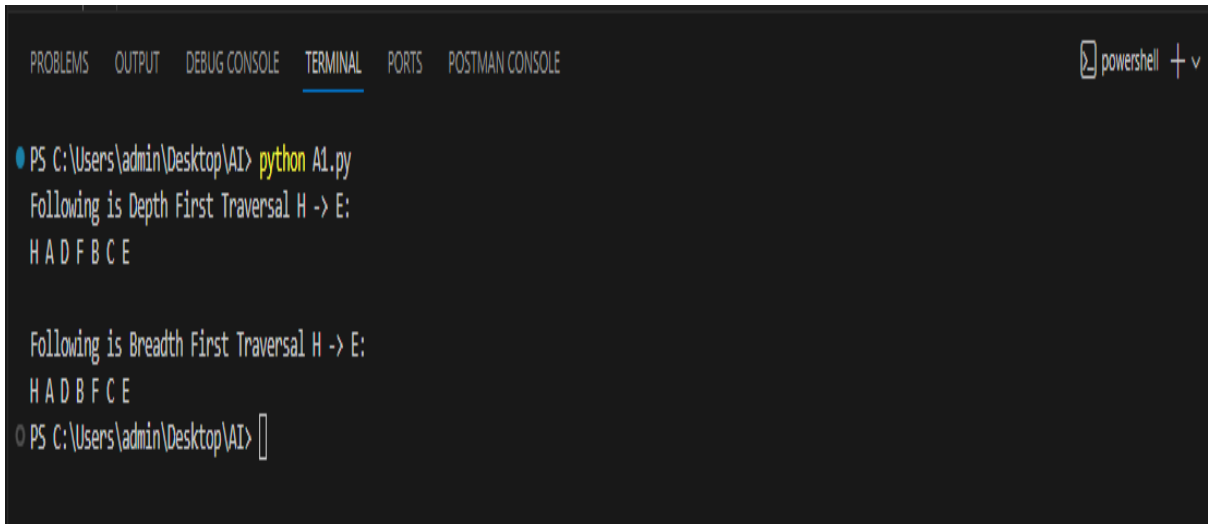
```
print("Following is Depth First Traversal H -> E:")
```

```
g.DFS('H', 'E')
```

```
print ("\n\nFollowing is Breadth First Traversal H -> E:")
```

```
g.BFS('H', 'E')
```

Output :



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE powershell + v

PS C:\Users\admin\Desktop\AI> python A1.py
Following is Depth First Traversal H -> E:
H A D F B C E

Following is Breadth First Traversal H -> E:
H A D B F C E
PS C:\Users\admin\Desktop\AI>
```